

SURVEY

Analyzing Brain Tumor Classification Techniques: A Comprehensive Survey

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ABSTRACT Classification of brain tumour is critical in medical image analysis and diagnosis as it aids in determining the affinity of cases to select treatment paths. This feature underscores the importance of accurate classification of brain tumours to predict their behavior, determine suitable management methods, and thus improve the performance of patients. The last few years have revolutionized medical picture grouping and brain tumour studies, thanks to machine learning tools, primarily CNNs. They demonstrate much potential for nearly extracting all features of interest from medical images and using them to classify tumours with high levels of accuracy. CNNs have complied with high accuracy rates, especially compared to the other forms of machine learning in this field. Due to the inherent ability to extract various aspects of the image along with the hierarchical detail of the picture, they are more suitable for use in operations like brain tumour classification. Similarly, the advancements in the field of natural language processing with transformer models have resulted in increased interest in the practical application of these types of models in image classification tasks. However, the application of ML models in the classification of brain tumours is still limited, therefore requiring continued research in this area. Therefore, the question about CNNs and Transformers' performance differences within the framework of the brain tumour classification becomes relatively more significant. This survey aims to provide recent contributions of related work in the domain and highlight the dynamism that has emerged due to such state-of-the-art methods.

INDEX TERMS Brain tumor classification, CNN, healthcare, medical image analysis, transformer.

I. INTRODUCTION

Detection of brain tumours is an important area of research and development in medical science as these tumors have a severe effect on the patient's life. Intracranial tumours are further categorized into primary and secondary; primary tumours which originate from brain tissues are either non-cancerous (benign) or cancerous (malignant) while the secondary ones originate from other areas of the body and are malignant than benign accounting to 28% of all intracranial

tumours. In America, the primary brain tumor detection rates for 2023 includes 67,440 [1] who will be diagnosed with non-malignant primary brain tumors and 26,940 with malignant brain tumors. More so, approximately between a hundred thousand and two hundred thousand patients are diagnosed with secondary brain tumor every year. Given the fact that brain tumors are the most common type of cancer, affecting one in 100 people, are aggressive and, with a mortality of nearly 100,000 cases annually, accurate diagnosis is crucial to enhance treatment and such figures define the necessity of timely diagnosis for better survival rates.

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For this survey, we selected papers published between 2018 and 2023 to capture the most recent advancements in brain tumor assessment using AI. We prioritized articles from high-impact journals to ensure the inclusion of rigorously peer-reviewed studies. Additionally, we considered significant conference papers from top-tier conferences to encompass cutting-edge research. Papers that reported lower results or did not show improvement over previous methods were excluded. This approach ensures our survey focuses on the most effective and innovative techniques in the field.

Medical imaging is an essential task that aids the clinicians and radiologists in diagnosing and defining the appropriate management of patients based on the type of brain tumor detected through a process known as Brain tumor classification (BTc). This domain has seen truly spectacular advancements within the past several years primarily thanks to the constructs of machine learning. This survey paper is intended to offer a brief review of brain tumor classification from recent studies, more specifically, whether the new trend of using transformers is superior to traditional CNNs in the classification of medical images such as brain tumor images.

In section of introduction, we outline the structure and content of this survey paper. Let us begin with the thorough description of other surveys relevant to BTc as depicted in [2]. This includes a comprehensive review of this paper which concerns itself with the application of machine learning methods, conventional methods, CNNs, and transformers among others in the context of brain tumor analysis [3], [4], [5]. To this end, the present paper is designed to begin from where previous studies have left off to offer a comprehensive understanding of the subject matter.

Brain tumour classification surveys: This section provides a review of the existing literature as well as current studies that has sought to analyze the efficiency of various supervised machine learning algorithms [6] for BTc. It also presents an overview of its development in this field and presents the authors' work on the application of machine learning algorithms to enhance diagnostics.

In the Comparative Intuition section, the authors devote an extensive discussion to comparing the results of machine learning-based approaches with the outcomes of the CNN-based methodologies [7] as well as discuss the differences between CNNs and transformer-based approaches [5]. Thus, the objective of research is to identify when and how much for specific parameters and scenarios. This paper discusses the applicability and drawbacks of each methodology in a bid to give an understanding how they can be effectively used for brain tumor classification under distinct conditions and specifications.

Next, in a detailed investigation of different datasets used in Brain Tumor analysis we proceed to discuss [8], [9], [10], [107]. This paper includes a synthesis of the relevant literature in terms of the features, advantages, and shortcomings of BRATS datasets used in the context of the BTc. Therefore,

through this case, we wish to give readers a detailed view of the existing datasets in the domain and help mitigate this issue by guiding their choice of datasets and their usage when conducting future studies.

Moving to the next section, we discuss the complexity of the errors we identified in the classification of the given brain tumor dataset and how they may be possible to solve. Further, we present some recommendations that are derived from our survey research that maybe useful for researchers and practitioners to follow in their pursuits in this area. In addition, we provide potential future research directions to facilitate academic advancement, identifying the best directions which still remain untapped and can be innovative. Thus, by outlining these difficulties and suggestions, we hope to contribute to the constant progression of both the ways in which to classify brain tumors and the overall development of medical imaging and diagnostics.

Lastly, in this section, we hereby provide the result of our survey alongside comparative analyses of the methodologies used in this domain as well as the various challenges that were highlighted in brain tumor classification. In this paper we presented also some considerations on the specific conclusions that have been carried out.

II. SURVEYS OF EXISTING LITERATURE ON BRAIN TUMOR CLASSIFICATION

The section provides a comprehensive literature that consists of prior research in the research area. The formal section of this work gives an overview of several works which addressed the problem of brain tumour classification [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41]. It discusses the CNNs, the transformer models and other machine learning techniques in light of response of brain tumours. Our aims in conducting this review are as follows: to present the overview of the ongoing progress in the given field, namely the BTc, and to indicate the directions, which have not been developed as thoroughly as can be wished.

A. MACHINE LEARNING FOR BRAIN TUMOR CLASSIFICATION

In this part of the study, we will discuss more on how various ml techniques are applied in analysis of brain tumours. This section describes various categories of ml techniques along with algorithms that can be applied for the classification of brain tumours. In the context of progression in the area of diagnostic imaging, we stress how crucial the enhancement of techniques is and how it contributes to increases in diagnostic capabilities and the utilization of therapeutic procedures. We also look at aspects of feature extraction and selection alongside various machine learning models that help to categorise the brain tumours with precision based on the imaging data. Out of this comprehensive analysis of using ML to classify brain

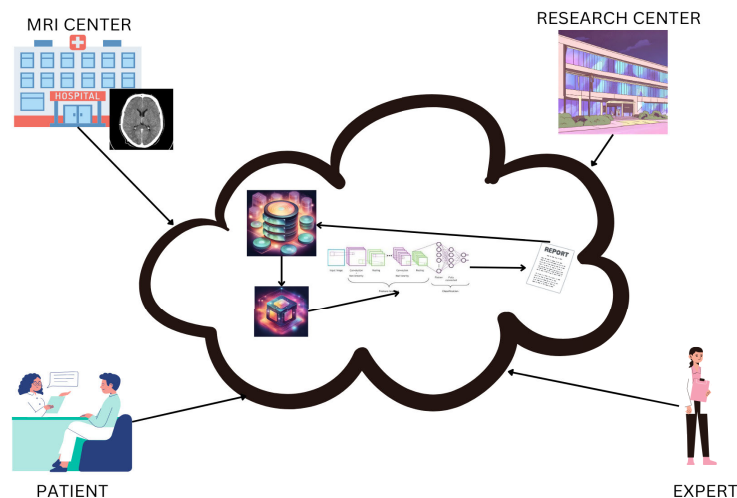


FIGURE 1. A pictorial representation that explains the real world healthcare and medical industry based methodology distinguishing and categorizing the real and virtual brain tumors with the help of traditional and AI approach.

tumours, we plan to identify the strengths, weaknesses, opportunities, and threats.

The computer aided diagnosis system is applied in the work, "A Multiclass Tumor Analysis of High-Grade Malignant Brain Tumors" [3] to classify the high-grade malignant brain tumours into various classes. The proposed strategy for feature selection enables the distinction between five subtypes of malignant brain tumours and employs a rather broad feature from six fields to maximize the amount of information that can be derived from the obtained results. The feature selection is established with a new approach known as CVM to facilitate selection of apt features from the feature set pool. Therefore, based on the above-mentioned selected features, three classifiers are trained and evaluated in terms of multi class classification accuracy. The results from the experiments are depicted to be of high performance. Remarkably, the proposed method, however, showed approximately 2%, 3% and 4% of increase in the overall accuracy compared to other algorithms such as ICA and GA with KNN, mSVM, and NN, respectively. Especially here, it is possible to point out that in the given tests, the highest effectiveness is shown by the NN classifier — 95% accuracy. 86% as depicted in the compact diagram below which proves competency and efficiency of the proposed methodology in classifying MBT into various classifications.

A new CADx system for staging stomach diseases: computer-aided diagnosis system for integrating the classification of diseases like gastric cancer, gastritis, and gastric ulcers [42]. This has arisen due to the low availability of medical data, something that is solved here through picture augmentation techniques using the Xception CNNs

with techniques such as depth wise separable convolution. Specifically, owing to the problem of feature visualisation in CNN, a class activation map (CAM) technique is applied to visualise the region of interest in the CNN, while the lesion site of abnormal images is superimposed on normal images to augment the dataset. Since anomalous photos are added to the dataset, the CNN training is improved with this augmentation. To evaluate the degree of the CADx system's effectiveness, the confusion matrix is employed as well as such specific and non-specific performance indicators as specificity, sensitivity, matrices. The recommendation with CAM effectively improves the dataset, as proven by boosting the original results further in form of F1score and AUC. In situations where the availability of raw data is minimal, the study contributes to the process of constructing an effective CADx that will help in the identification of potential stomach diseases using images obtained from gastroscopy.

A hybrid machine learning system [43] Texture, intensity, and form feature extraction, using a MAP-based firefly algorithm, and data. The suggested approach achieves a dependable accuracy of 99.2% for brain strokes and 88.3% for brain tumours, with three different locations discovered throughout the tumour identification procedure and MRI pictures of brain tumours and strokes separated into four groups. The best F-score of 0.91 and the lowest False Positive Rate (FPR) of 0.06 for brain tumour classification and an F-score of 0.99 and 0.0 FPR for brain stroke classification are achieved by the hybridised support vector-based random forest classifier (HSVFC), demonstrating better performance. In addition, HSVFC performs better than SVM, FFNN, DC, ResNet-18, and KNN classifiers; its maximum mean accuracy for classifying brain tumours is 76.55%, while its

maximum mean accuracy for classifying brain strokes is 98.17%.

The study [44] taking advantage of its multicoloured tissue depiction to improve comprehension of the problem. The firefly algorithm is utilised to optimise clustering processes. In order to close the research gap, the study emphasises the necessity of comprehensive details regarding the hyperspectral imaging dataset, a comparative analysis with alternative tumour localization methods, evaluation on a variety of datasets, and an examination of the interpretability of the results. The outcomes demonstrate the efficacy of the suggested method, outperforming previous techniques in tumour localization accuracy with accuracy of 96.47%, sensitivity of 96.32%, and specificity of 98.24%.

MRI-Based BTc [45]. Using DNN that have already been trained. The best three deep features are then integrated to form an ensemble after these features are assessed by a number of machine learning classifiers. The study makes use of three publicly available brain MRI datasets for assessment, but it also identifies a number of research gaps, such as the absence of comparison with DL models tailored for medical image analysis, the lack of detailed dataset information, and the requirement for the proposed ensemble of deep features to be interpretable and explainable. While the experimental results show promising improvements in performance, the relevance and efficacy of the results cannot be fully evaluated due to the absence of precise metrics and comparisons with other sophisticated techniques.

B. CNNs FOR BRAIN TUMOR CLASSIFICATION

We present a thorough the use of CNNs in brain tumour analysis in this section. This section includes a detailed explanation of the many CNN designs and techniques used in brain tumour classification. We examine the development of CNN techniques in medical imaging, emphasising their importance in improving diagnostic performance and guiding therapeutic methods. We also examine how features are extracted and selected within CNN frameworks and how various CNN models contribute to the precise classification of brain tumours using imaging data. We hope to clarify the benefits, drawbacks, and future directions of using CNNs for brain tumour classification by this thorough investigation.

The paper [46] presents a novel approach to brain MRI classification. The method aims to leverage unlabeled data to learn meaningful features without requiring human annotations, addressing the challenge of limited labeled data availability in brain MRI images. The self-supervised learning procedure involves data augmentation and the creation of model for feature extraction. The pre-trained model is then utilized for supervised learning and MRI image categorization as an encoder. The study investigates the efficacy of contrastive loss-based pre-training, as well as the impact of dataset diversity and augmentation types on performance. However, the research gap includes the lack of information on the unlabeled picture dataset used, the

need for detailed explanations of data augmentation methods, and further exploration of pre-training configurations and contrastive loss hyperparameters' effects on classification performance. Experimental results demonstrate that contrastive loss-based self-supervised pre-training outperforms random or ImageNet initialization, and the diversity of images in the pre-training dataset positively affects contrastive learning effectiveness. Additionally, the type of data augmentation employed for creating positive and negative examples influences final classification accuracy.

The problem of segmenting brain tumours in multi-modal MRI data with missing modalities is addressed in the study [47]. It suggests a Dual Disentanglement Network (D2-Net) that specifically takes into account connections between knowledge particular to tumours and information specific to other modalities. The MD-Stage and the TD-Stage. Research gaps do exist, nonetheless, such as the absence of ablation studies and performance comparisons with currently available methods to assess the unique contributions of MD-Stage and TD-Stage. Furthermore, additional investigation is required to evaluate the effects of varying degrees of missing modalities and the applicability of D2-Net to various datasets related to brain tumours. Based on BraTS-2018 dataset experiments, D2-Net performs better than other methods in brain tumour segmentation when modalities are lacking. The technique produces more dependable and precise segmentation results by efficiently capturing correlations between modalities and tumour sites. The suggested method's efficacy is further increased by affinity-guided dense tumor-region information.

The paper [48]. Using three BraTS datasets. K-fold cross-validation is employed to evaluate the model's generalizability. The results demonstrate that the TD-CNN-LSTM network achieves a test accuracy of 98.90%, outperforming the 3D CNN model. However, research gaps include the lack of comparison with existing cutting-edge deep learning models, exploration of various layer configurations and hyperparameters' effects on model functionality. Nonetheless, the study underscores the effectiveness of the TD-CNN-LSTM model in analyzing all four MRI sequences simultaneously for brain tumor classification, highlighting its potential in clinical settings.

A paper [52]. For the purpose of brain tumour detection, a variety of transfer learning-based deep learning methods, like, ResNet50, InceptionResNetV2, InceptionV3, Xception, and DenseNet201, In order to improve the efficacy of the early brain tumour detection process, the study looks into transfer learning using seven deep learning approaches and conventional classifiers. The findings show that the optimal model with a 10-fold cross-validation attains an accuracy of 99.39%, and different feature extractor and classifier combinations perform better on a variety of measures. The learning curves that are displayed offer valuable insights into the training procedure and performance patterns of combinations that attain high accuracy, helping to choose the most suitable methods for brain tumour detection.

TABLE 1. Comparative analysis of all methods and discussion on advantages and disadvantages of machine learning, CNNs and transformers techniques.

Approach	Main Contributions	Advantages	Disadvantages
Machine Learning	Novel methodology for multi-class classification of malignant brain tumors [3]	High classification accuracies (up to 95.86%)	Limited by the need for manually engineered features
Machine Learning	Enhancement of CADx system for gastric disorder classification through image augmentation [42]	Improved performance in F1 score and AUC	Lack of detailed dataset information
Machine Learning	Hybrid machine learning system for brain abnormality classification in MRI images [43]	Reliable accuracy in brain tumor and stroke classification	Requires further comparison with other techniques
Machine Learning	Optimization of clustering processes for tumor localization in brain using hyperspectral imaging [44]	Higher accuracy and specificity	Lack of evaluation on diverse datasets
CNNs	Introduction of self-supervised pre-training with contrastive loss for brain MRI classification [46]	High validation and test accuracies above 97%	Lack of comparison with existing techniques
CNNs	Proposal of Dual Disentanglement Network for brain tumor segmentation in multi-modal MRI data [47]	More reliable and accurate segmentation results	Lack of performance comparison with existing techniques
CNNs	Development of TimeDistributed-CNN-LSTM hybrid model for brain tumor classification using 3D MRI sequences [48]	Achieves high test accuracy of 98.90%	Lack of comparison with cutting-edge deep learning models
CNNs	Investigation of deep transfer learning models for brain tumor detection using 2D MR images [45]	Superior performance in accuracy and F1 score	Lack of detailed dataset information
Transformers	Evaluation of pretrained and fine-tuned Vision Transformer (ViT) models for brain tumor classification [49]	Consistently high validation and test accuracies above 97%	Lack of exploration of other ViT variants
Transformers	Introduction of Linear Complexity Data-Efficient Image Transformer (LCDEiT) for brain tumor classification [50]	High classification accuracy and F1-scores with linear computational complexity	Lack of comparison with existing techniques
Transformers	Proposal of Voting Based Semi-Supervised Bayesian Ensemble Attention Mechanism (VS-BEAM) for brain tumor segmentation and classification [51]	Impressive accuracy of 98.91% on test dataset	Lack of detailed dataset information and comparison with other techniques

A paper [53]. The method uses deep transfer learning using imbalanced data to fine-tune the Densenet201. The average pool layer is used to extract features, and two feature selection methods are suggested. Although the suggested approach obtains over 95% accuracy on the BRATS2018 and BRATS2019 datasets, comprehensive details about the datasets' properties, histories, and pretreatment processes are absent. It is imperative to conduct additional research on the ability of feature selection techniques, such as EKbHFV and MGA, to pick discriminative features and to examine the effects of varying threshold settings and hyperparameters on overall performance. The study should also look into the method's potential for real-world clinical applications, as well as its resilience and generalizability on external datasets. The literature study, however, emphasises how the suggested methodology outperforms the other neural nets in the comparison, underscoring its usefulness and effectiveness for multiclass brain tumour classification.

The paper [54] investigates the application of Machine Learning models for brain tumor detection. Highlighting the hallmark of brain tumors as the collection of cells with unrestricted development, the authors discuss previous research utilizing Machine Learning models trained on domain-specific datasets for tumor identification, presenting their own experimental findings. While the report expands on past work in brain tumor detection using Machine Learning models, it lacks specification of a particular research gap being addressed. However, utilizing the RESTNET 5.0 with Transfer Learning Model, the authors achieve an accuracy of 96% in recognizing Glioma tumors, Meningioma tumors, and Pituitary tumors. They note that this model outperformed six different Machine Learning algorithms tested on their dataset, demonstrating its effectiveness in brain tumor classification.

Early brain tumour diagnosis utilising 2D magnetic resonance (MR) pictures, a study [55]” was conducted. Although the study uses deep learning-based classification

techniques on 2D MR images to meet the need for early brain tumour diagnosis, it does not clearly identify the research gap that needs to be filled. But the real issue is figuring out which combination of deep learning feature extractors and classifiers would improve brain tumour detection systems' efficiency. The best-performing model obtains a 10-fold cross-validation accuracy of 99.39%, according to the results.

Using Internet of Medical Things (IoMT) technology, [56] aims to create a CAD system. The suggested system extracts from MRI scans and classifies them into three categories of tumours: gliomas, meningiomas, and pituitary. It does this by using deep learning techniques, namely transfer learning using GoogleNet. The suggested model's assessment criteria such F1 score, accuracy, and specificity. By utilising IoMT technology and deep learning to address the research gap in early tumour detection and classification, the study outperforms previous models with respect to accuracy, specificity, and F1 score. The suggested model works better than the current ones, as evidenced by the experimental findings, which show an accuracy of 96.9%, specificity of 96.6%, and F1 score of 0.969. These findings demonstrate the model's practical applicability in the early identification and categorization of brain tumours with the use of IoMT technology.

C. TRANSFORMERS FOR BRAIN TUMOR CLASSIFICATION

We take a close look at how Transformers are used in the field of brain tumour analysis in this section. This section includes a detailed explanation of the many Transformer topologies and techniques used in brain tumour categorization. We examine the development of Transformer techniques in medical imaging, emphasising their importance in improving diagnostic capacities and guiding therapeutic methods. We also examine how features are extracted and selected inside Transformer frameworks and how various Transformer models contribute to the precise classification of brain tumours using imaging data. We hope to clarify the benefits, drawbacks, and future directions of using Transformers for brain tumour classification with this thorough analysis.

The performance of pretrained and refined Vision Transformer (ViT). Evaluations show that both individual ViT models and an ensemble approach regularly produce validation and test accuracies above 97%. Robust performance is shown via hyperparameter adjustment in various settings, with the Adadelta optimizer surpassing the others at 384×384 resolution. Higher resolutions often result in better performance, according to test results, which also indicate that models with the highest validation accuracy perform better. The ViT ensemble outperforms other studies, obtaining an overall test accuracy of 98.7%, according to comparisons. More misclassifications of meningiomas are seen, most likely as a result of feature overlap. The study demonstrates the superiority of pretrained and fine-tuned ViT models, outperforming earlier CNN-based methods. The best results came from fine-tuning every model parameter,

even with computing loads; this could lead to more research into ViT variations and greater resolutions. Other datasets can be used with cross-validated models, and different ViT topologies may be investigated in subsequent studies. Code and models can be accessed upon request. [49].

In order to solve problems with deep learning models in brain tumour classification, the paper LCDEiT focuses on data efficiency and computational complexity. To address these issues, linear-complexity data-efficient scans transformer called LCDEiT is presented, which combines a teacher-student approach and an external attention mechanism. On benchmark Figshare and BraTS-21 dataset, LCDEiT demonstrates high classification F1-scores and accuracy, indicating its design for medical scan jobs demanding rapid calculations and limited data. The average accuracy and F1-scores of the experimental findings are 98.11%, 97.86%, 93.69%, and 93.68%, respectively. These results point to the potential influence of the system on medical diagnosis, where quick analysis and sparse data availability are crucial. By increasing data efficiency and achieving linear computing efficiency, the suggested model fills in the holes in current models [50].

The tumor extraction method of the VS-BEAM proposed in this paper applies a convolutional autoencoder network results to tumor extraction and enjoys high accuracy in multi-class tumor classification. However, there are areas that future research can improve as follows: Failure to provide details about the data set used in the study, Comparison of the method with other techniques for a more general result, Evaluation of how robust the models developed are and if they can be applied to other data sets. Nevertheless, the qualitative nature of the research and an insight into the potential areas of clinical application of the VS-BEAM decision-making model should be further explored. Still, as seen in each of the categories above, there are some gaps, yet the study records a promising throughput estimation accuracy of 98. Validation on a test dataset including 264 brain tumor site note. One hundred for each protocol with average sensitivity of VS-BEAM reaching 91% [51].

This paper thus reviews state of the art approaches involving transfer learning, CNNs, CapsNets, and vision transformers (ViTs) implemented for MRI data-based brain tumour segmentation as well as classification. It identifies several directions for further work to be done and discusses several research directions, including the development of an algorithm that can simultaneously estimate the probabilities of multi-grade samples, segmentation of tumour regions and classification of tumour samples. The survey also outlines the areas to address in terms of methods for how to generate better feature maps to augment the current segmentation and classification through CNN-based architectures, analysing CapsNets and ViTs in diagnosing brain tumours. The paper also suggests that in order to increase the network performance it is necessary to look into network topologies in the third dimension, manage with model uncertainty, look at different activation functions in deeper details and make

sure that the feature maps are in high resolution to reduce the imbalance between training and testing data set; one should also use pre-trained networks and complex augmentation strategies [57].

In this study, one of the image recognition approaches known as Vision Transformer (ViT) is discussed. Essentially, it is a variant of the normal Transformer architecture, which has not been modified for image processing. Some of the architecture's defining features make it very simple and highly effective when trained on massive data sets. It receives an image as pieces and alters it with a typical Transformer encoder [58].

In this research presented novel hybrid D.L model for labeling the content into the BT categorization. As a result of the integration of the two forms of networks, namely the transformer and CNNs in the recommended model, the following capabilities were achieved. Moreover, there are the FFM and IMM used in our study, which enhances feature representation capacity, to gain more desirable outcomes. In the experiments, it was proven that the proposed model can be instrumental to help the radiologist to make the correct classification decision for BTc [59].

To achieve this, a new CBMIR method is proposed in this work in order to make retrievals of images of brain tumours from large datasets of MRI as efficient as possible. In the procedure, it employs Walsh-Hadamard transform, Gabor filtering, and executing the noise removal steps to distillate the features. Minkowski distance metric is used for computation in Fuzzy C-Means and for retrieval purpose. Analyzing this result in comparison to other currently applied technologies like wavelet, Gabor and Hough transforms, it can be concluded that the suggested CBMIR system is more effective in diagnosing brain tumours from MRI. In addition, it also manifested the efficiency also in aspect of time and therefore is capable of efficient real time applications. Potential areas for future work therefore are: the degree of flexibility with regards to all types of brain tumours, the improvements in MRI picture resolution and/or clarity as well as exploring the issue of scalability and applicability to larger and more diverse MRI sets [60].

D. OTHER MEDICAL IMAGE CLASSIFICATIONS METHODS

Stereo imaging technology has revolutionized medical diagnostics [61] changes such as low resolution (LR) and blurry output images. Traditional super-resolution (SR) techniques often involve complex blocks and loss functions, leading to high system complexity. To address this, we propose a combined channel and spatial attention block, integrated with a parallax attention module (PAM), for enhancing endoscopic image resolution. Trained on the da Vinci dataset, our model significantly improves PSNR and SSIM, suggesting enhanced diagnostic and treatment accuracy for endoscopic images. This advancement underscores the potential of stereo imaging in providing high-resolution medical images critical for accurate diagnosis and effective treatment planning across

various medical fields, including cardiology, gynecology, and gastrointestinal surgery.

Body composition (BC) is crucial in assessing the risk of type 2 diabetes and cardiovascular disease, with computed tomography (CT) being a valuable imaging method for BC analysis [62]. However, manual segmentation of CT images is labor-intensive and subjective. This study aims to develop and evaluate fully automated segmentation techniques using a 3-slice CT imaging protocol, which includes slices at the liver, abdomen, and thigh levels for detailed analysis of various tissues and organs. Utilizing over 4000 CT subjects from the SCAPIS and IGT cohorts, four convolutional neural network-based architectures ResUNET, UNET++, Ghost-UNET, and Ghost-UNET++ were trained and evaluated. These models aimed to segment the liver, spleen, skeletal muscle, bone marrow, cortical bone, and various adipose tissue depots, including visceral (VAT), intraperitoneal (IPAT), retroperitoneal (RPAT), subcutaneous (SAT), deep (DSAT), and superficial SAT (SSAT), as well as intermuscular adipose tissue (IMAT). The models underwent tenfold cross-validation and testing for each target. Results indicated strong performance across all models, with Dice scores showing UNET++ as the top performer, achieving 0.981 (0.927–0.996), while Ghost-UNET++ offered competitive performance with better computational efficiency. This demonstrates that fully automated segmentation can significantly streamline BC analysis, providing objective results efficiently in large-scale studies.

While they were adapted for word-based challenge during its inception, the latter transformer-based designs have also been shown to handle complex high dimensional massive scale neuroimaging data very well. Thanks to this mechanism, challenges such as the segmentation and classification of brain tumors are made easier since accurate features can be derived. Transformers have been proven to be efficient in improving diagnostic processes as studies have indicated that they are more accurate and less computational time consuming in comparison to standard convolutional networks [63]. However, there are still plenty of research being done on such topics as hardware expenses, model interpretability, and data insufficiency. Finally, it can be envisaged that the increasing implementation of transformers for the analysis of brain MRI will benefit the early diagnosis and management of obstruction due to brain cancers.

Therefore, the proposed saliency-aware SR model [64], is advocated as one for relieving the limitations in stereo imaging of endoscopy, especially concerning yielding high-quality images. Although stereo imaging can convey depth information during the endoscopic procedure, it often gives low-resolution images that may challenge diagnosis. Our model is intended to present much of the valuable structural information well preserved while improving the image quality with a newly designed feature extraction and cross-view interaction module. Our model performs better in PSNR and SSIM scores than the existing techniques, which include E-SEVSR, DCSSRNet, and CCSBESR, on variable datasets.

Not only do these enhanced image quality and clarity allow surgeons to perform difficult operations with greater ease, but they are likely to create new limits for the medical imaging technologies themselves, potentially enabling completely new discoveries and interventions in endoscopic diagnostics and therapy.

III. COMPARATIVE STUDY

The current section presents a comprehensive review and discussion on various methodologies available in the literature concerning brain tumor classification. It starts with a comparison of conventional ML methodologies to CNNs to establish how the last is superior, though with its own constraints when analyzing image data and more so when it comes to brain tumor identification. The section then proceeds to compare CNNs with another revolutionary architecture in deep learning-known as the Transformers-following a brief explanation of the differences between the two and their applicability to image data. Therefore, in this section, the review hopes to draw comparison and understanding of other methods and their characteristics and likely impacts towards further development of brain tumor classification techniques.

A. OVERALL COMPARISON

The responses collected in the course of the survey study presented in this section aim to outline several types of paper, as well as compare the approaches and conclusions assumed while accomplishing them, along with the possible implications. It tries to make the readers understand the superficial similarities and differences, the loops that occur, and the reason behind it all, which gives them a clear vision of the literature. For each paper, the presented results and the specific contributions are properly discussed, and major findings are emphasised.

The Table 2 presents a overview of research studies in the brain tumor detection, classification, and segmentation, focusing on the approach of machine learning (ML), convolutional neural networks (CNNs), and transformers. Each study is characterized by its approach, dataset used, accuracy achieved.

For instance, various studies explore different methodologies, such as SVM and other machine learning approaches and decision trees, CNN architectures, and transformer models, to tackle specific tasks related to brain tumor analysis. Some studies propose novel architectures or incorporate attention mechanisms and adversarial training to enhance model performance and robustness.

However, despite the advancements, there exist significant research gaps that need further exploration. These gaps include the need for improved accuracy through the exploration of CNNs in certain studies, investigation into model robustness across diverse datasets, limited exploration of hyperparameter tuning and model interpretability, and the necessity for standardized evaluation metrics and benchmark datasets.

Overall, the synthesis of these studies provides valuable insights into the current state of research in brain tumor analysis, highlighting both the progress made and the avenues for future investigation to advance the field.

Figure 2 presents a histogram illustrating the accuracy of various papers in the field of brain tumor analysis. Each bar represents the accuracy percentage achieved by a specific research study. The accuracy percentages are annotated on the bars themselves, providing a clear visualization of the distribution of accuracies across different approaches and methodologies explored in the literature.

Figure 10 depicts a histogram showing the distribution of accuracies achieved by machine learning (ML) approach models in brain tumor analysis. The x-axis represents accuracy percentages, while the y-axis indicates the frequency of models falling within each accuracy range.

Figure 4 illustrates a histogram presenting the accuracies of convolutional neural networks (CNNs) approach models in brain tumor analysis. The histogram displays the distribution of accuracies across different CNN architectures used in the studies.

Figure 5 shows a histogram representing the accuracies achieved by transformer-based approach models in brain tumor analysis. The graph provides insights into the distribution of accuracies among various transformer architectures explored in the literature.

Figure 6 displays a histogram demonstrating the average accuracy of all approaches, including machine learning, CNNs, and transformers, in brain tumor analysis. The histogram presents a combined view of the accuracies achieved by different approaches, providing an overall perspective on the performance of the models in the field.

B. COMPARATIVE STUDY BETWEEN MACHINE LEARNING AND CNNs

Machine learning and CNNs represent two prominent methodologies utilized in various domains, including brain tumor classification. While both approaches aim to enhance diagnostic capabilities and inform treatment strategies, they differ significantly in their underlying principles, capabilities, and applications. In this comparative study, we analyze the strengths, limitations, and potential advancements of machine learning and CNNs in the context of brain tumor analysis.

1) Feature Representation and Extraction:

- **Machine Learning:** Traditional machine learning techniques rely on handcrafted feature extraction methods, where domain knowledge is used to design features that best represent the underlying. These features fed into machine learning algorithms [3].
- **CNNs:** Conversely, CNNs automatically extract hierarchical feature representations from unprocessed input data. CNNs are highly suitable for tasks like image classification because they can extract complicated spatial and temporal patterns

TABLE 2. A comparative analysis of machine learning, CNNs, and transformers in brain tumor assessment.

Sr.No.	Paper Topic	M.L.	CNNs	Transformers	DataSet	Accuracy	Description	Research Gap
1	[65]	✓			BRATS-2019, BRATS-2021 [65]	85%	Compares SVM, KNN, and Decision Trees for tumor detection	Further exploration of CNNs for improved accuracy
2	[66]		✓		MRI SCAN [66]	92%	Proposes a CNN architecture for tumor classification	Investigating ensemble methods to enhance model performance
3	[67]			✓	AANLIB [67]	87%	Utilizes transformers for tumor classification	Limited exploration of hyperparameter tuning for optimal performance
4	[68]	✓			MRI SCANS [68]	88%	Evaluates SVM, Random Forest, and Decision Trees for segmentation	Lack of investigation into model robustness across diverse datasets
5	[69]		✓		BRATS-2019, BRATS-2021 [69]	94%	Introduces CNNs with data augmentation techniques	Limited discussion on model interpretability and explainability
6	[70]			✓	MULTIGRADE [70]	90%	Incorporates attention mechanisms in transformer models	Exploration of multi-modal fusion for improved classification performance
7	[71]	✓			MRI SCANS [71]	Varies	Reviews various ML techniques for tumor detection and classification	Need for standardized evaluation metrics and benchmark datasets
8	[72]		✓		BRATS-2019, BRATS-2021 [72]	Varies	Surveys CNN architectures for tumor detection	Exploration of transfer learning strategies for small dataset scenarios
9	[73]			✓	MULTIGRADE [73]	89%	Proposes transformer models for tumor segmentation	Investigation into domain adaptation techniques for model generalization
10	[74]	✓	✓		MRI SCANS [74]	93%	Integrates both ML and CNN approaches	Analysis of interpretability and clinical relevance of model predictions
11	[75]	✓	✓		AANLIB, MULTIGRADE [75]	91%	Explores transfer learning with pre-trained CNNs	Investigation into domain adaptation techniques for heterogeneous datasets
12	[76]		✓		BRATS-2019, BRATS-2021 [76]	96%	Utilizes attention mechanisms in CNNs	Further investigation into model explainability and feature importance analysis
13	[77]	✓			MRI SCANS, MULTIGRADE [77]	87%	Applies RL techniques for tumor segmentation	Exploration of multi-task learning frameworks for improved segmentation accuracy
14	[78]	✓	✓		BRATS-2019, BRATS-2021 [78]	90%	Incorporates adversarial training for model robustness	Investigation into model generalization across different imaging modalities
15	[79]	✓	✓		MRI SCANS [79]	88%	Utilizes GANs for tumor synthesis and classification	Analysis of data augmentation strategies for small datasets
16	[80]		✓		MRI SCAN [80]	85%	Adapts GCNs for tumor graph classification	Investigation into graph representation learning techniques for improved performance
17	[81]	✓			MRI SCANS [81]	86%	Utilizes sparse coding techniques for subtype classification	Exploration of semi-supervised learning methods for enhanced model performance
18	[82]	✓	✓		BRATS-2019, BRATS-2021 [82]	92%	Applies transfer learning with genetic algorithms	Investigation into model interpretability and clinical feasibility
19	[83]		✓		AANLIB [83]	91%	Adopts capsule networks for tumor classification	Exploration of capsule network architectures for improved scalability
20	[84]		✓		MULTIGRADE [84],	88%	Incorporates uncertainty estimation in CNN-based segmentation	Investigation into uncertainty calibration techniques for reliable model predictions
21	[85]	✓			MULTIGRADE, AANLIB [85]	84%	Utilizes fuzzy logic for severity classification	Analysis of ensemble methods for improved classification robustness
22	[86]	✓			BRATS-2019, BRATS-2021, AANLIB [86]	89%	Integrates SVM and Decision Trees for tumor detection	Investigation into feature selection techniques for improved model efficiency

TABLE 2. (Continued.) A comparative analysis of machine learning, CNNs, and transformers in brain tumor assessment.

23	[87]			✓	MRI SCAN [87]	90%	Incorporates attention mechanisms in GNNs for segmentation	Exploration of graph-based data augmentation strategies for improved model generalization
24	[88]	✓	✓		MRI SCANS [88]	Varies	Investigates ensemble techniques for classification	Analysis of diversity-enhancing strategies for ensemble models
25	[89]		✓		MULTIGRADE, MRI SCANS [89]	87%	Utilizes VAEs and GANs for tumor segmentation	Investigation into uncertainty quantification techniques for reliable segmentation
26	[90]	✓			BRATS-2019, BRATS-2021 [90]	88%	Explores self-supervised learning with contrastive methods	Analysis of self-supervised pre-training strategies for model initialization
27	[91]	✓	✓		MRI SCANS [91]	90%	Investigates adversarial attacks on segmentation models	Exploration of defense mechanisms against adversarial attacks in medical imaging
28	[92]	✓	✓		BRATS-2019, BRATS-2021 [92]	89%	Addresses limited annotation challenges with semi-supervised learning	Investigation into active learning strategies for efficient annotation acquisition
29	[93]	✓			MRI SCANS [93]	86%	Explores meta-learning techniques for few-shot learning scenarios	Investigation into meta-learning algorithms robustness to dataset shifts
30	[94]	✓			MULTIGRADE, MRI SCANS [94]	87%	Utilizes metric learning for tumor segmentation	Analysis of metric learning loss functions for improved segmentation performance

from pictures using convolutional layers, pooling layers, and nonlinear activation functions [47].

2) Data Complexity and Scalability:

- **Machine Learning:** While traditional ML techniques can handle a wide range of data types and formats, they may struggle with high-dimensional features, such as medical scans, due to the curse of dimensionality. Additionally, feature engineering can be time taking and may require domain-specific expertise [43].
- **CNNs:** CNNs excel at processing high-dimensional features, such as MRIs, without the need for manual feature engineering. They are making them highly scalable and adaptable to diverse datasets [55].

3) Interpretability and Explainability:

- **Machine Learning:** Because the features used for categorization are explicitly set by domain experts, traditional machine learning models [95] are frequently easier to understand and comprehend than convolutional neural networks (CNNs). This openness can be very important in medical settings where decisions must be understood and defended. [46].
- **CNNs:** CNNs often regarded as uninterruptible models, meaning that it can be hard to interpret how they arrive at their results. While techniques such as activation maps and gradient-based methods [96] can provide some insight into

model decision-making, CNNs generally lack the interpretability [97] of traditional ML models [48].

4) Performance and Accuracy:

- **Machine Learning:** The performance of traditional ML models heavily depends on the quality of handcrafted features and the choice of classifier. While ML models can achieve high accuracy with carefully engineered features and optimized hyperparameters, they may struggle [98] with complex, high-dimensional data [52].
- **CNNs:** In image for classification tasks, including the classification of brain tumours [99], CNNs shown state-of-the-art results. CNNs can detect complex patterns in medical pictures [100] by automatically learning hierarchical representations, which improves accuracy over typical machine learning techniques [55].

5) Data Efficiency and Generalization:

- **Machine Learning:** Traditional ML models may require large chunk of labeled data to achieve optimal performance, especially when using complex feature representations. Generalization to unseen data can be challenging if the model overfits to the training data [54].
- **CNNs:** CNNs have shown to be data-efficient learners, capable of generalizing well to unseen data even with limited labeled examples [75]. Transfer learning methods further enhance generalisation performance [46].

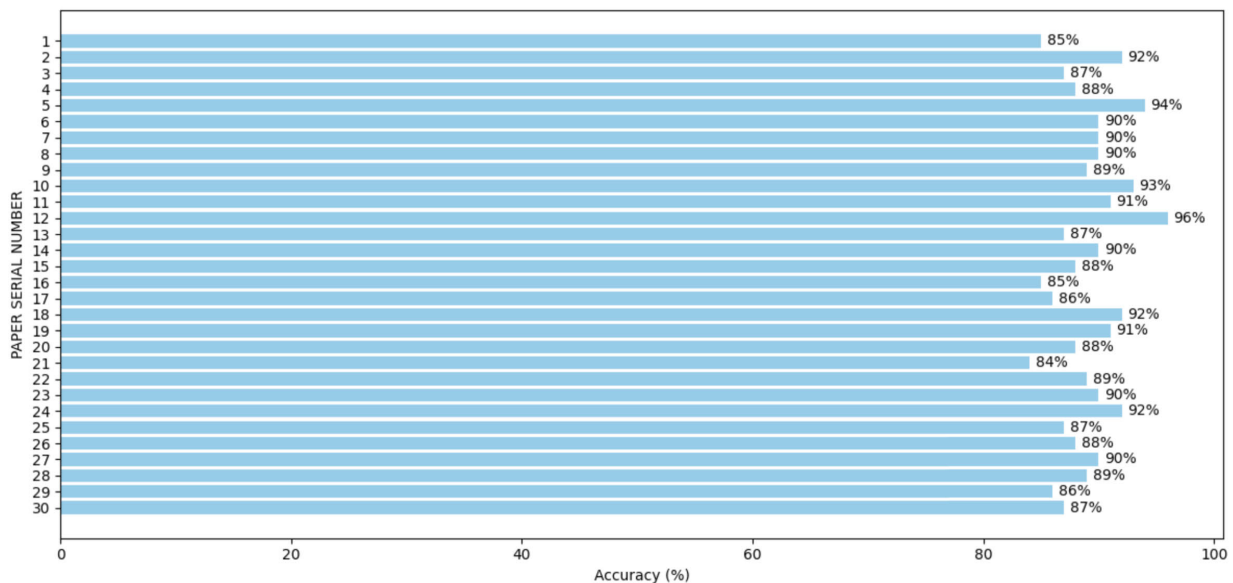


FIGURE 2. Compare the accuracy of different brain tumor classification algorithms, where the y-axis represents the serial numbers and the x-axis represents their respective accuracies from Table 2.

C. COMPARATIVE STUDY BETWEEN CNNs AND TRANSFORMERS

Transformers and Convolutional Neural Networks (CNNs) are two potent methods used in the field of brain tumour analysis. In this paper, we compare the efficaciousness, drawbacks, and future directions of CNNs and Transformers in the precise classification of brain tumours from imaging data.

1) Feature Representation and Extraction:

- **CNNs:** CNNs learn hierarchical feature [101] representations directly from raw input data. The use of cnn layers, pooling layers, and nonlinear activation functions, CNNs can extract complex spatial and temporal patterns from images [47].
- **Transformers:** Transformers process images by treating them as sequences of patches and utilizing self-attention mechanisms to capture long range non-linear dependencies. They excel in capturing global relationships within images but may require large amounts of data for effective training due to their reliance on self-attention [58].

2) Data Complexity and Scalability:

- **CNNs:** CNNs are an excellent choice if you want to process high-dimensional data without requiring manual feature engineering, like MRI scans [102]. They are extremely scalable and versatile to a variety of datasets since they can automatically extract pertinent features [103] from raw input data [55].
- **Transformers:** Transformers may require large amounts of data for effective training due to their reliance on self-attention mechanisms [104].

While they excel in capturing global relationships within images, they may struggle with limited data availability and computational complexity [50].

3) Interpretability and Explainability:

- **CNNs:** CNNs are often regarded as uninterruptible models, making it uninterpreted how they arrive at their predictions. While techniques such as activation maps can provide some insight, CNNs generally lack the interpretability of traditional machine learning models [48].
- **Transformers:** Through self-attention mechanisms, transformers provide interpretability, enabling users to see the attention weights and determine which areas of the image are most important for the predictions. In medical contexts where decisions must be understood and defended, this openness might be beneficial [58].

4) Performance and Accuracy:

- **CNNs:** CNNs have demonstrated state-of-the-art performance in a range of image classification applications, including the categorization of brain tumours. In contrast to traditional machine learning methods, CNNs can detect complex patterns [105] in medical pictures by automatically learning hierarchical representations, which improves accuracy [55].
- **Transformers:** Transformers have shown promising results in image classification [104] tasks, including brain tumor classification. While they may require large amounts of data for effective training, transformers can capture global

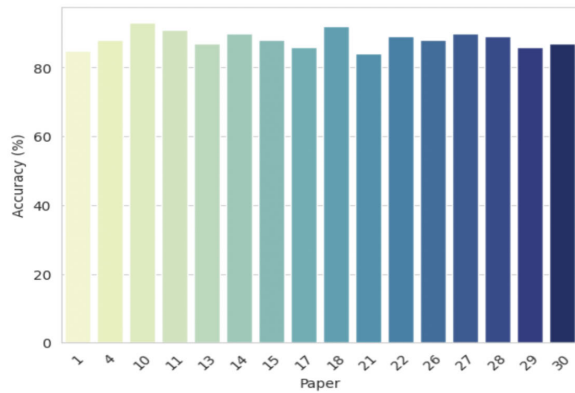


FIGURE 3. Compare the accuracy of different ml based brain tumor classification algorithms, where the y-axis represents the accuracy and the x-axis represents the serial numbers from Table 2.

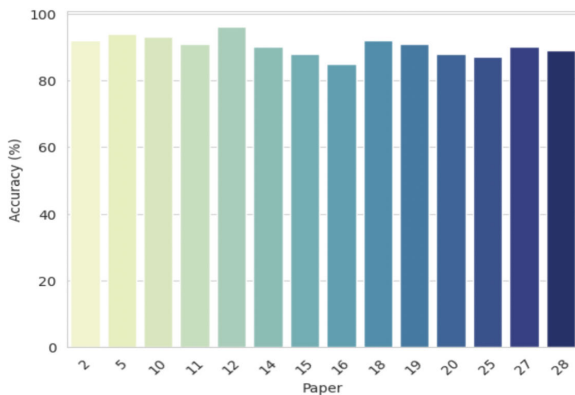


FIGURE 4. Compare the accuracy of different cnn based brain tumor classification algorithms, where the y-axis represents the accuracy and the x-axis represents the serial numbers from Table 2.

relationships within images, leading to competitive performance in classification tasks [49].

5) Data Efficiency and Generalization:

- **CNNs:** CNNs have demonstrated data-efficient learning capabilities, capable of generalizing well to unseen data even with limited labeled examples [106]. Transfer learning techniques further improve generalization performance, making CNNs well-suited for medical imaging tasks with limited labeled data [55].
- **Transformers:** Transformers may require larger labeled data for effective training due to their reliance on self-attention mechanisms [5]. However, they have shown promising results in generalization to unseen data, especially when pre-trained on large-scale datasets [50].

D. RESEARCH GAP IN BRAIN TUMOR TECHNIQUES

The comparative analysis of recent studies in brain tumor assessment reveals several research gaps that could be addressed to enhance the performance and applicability of machine learning (ML), convolutional neural

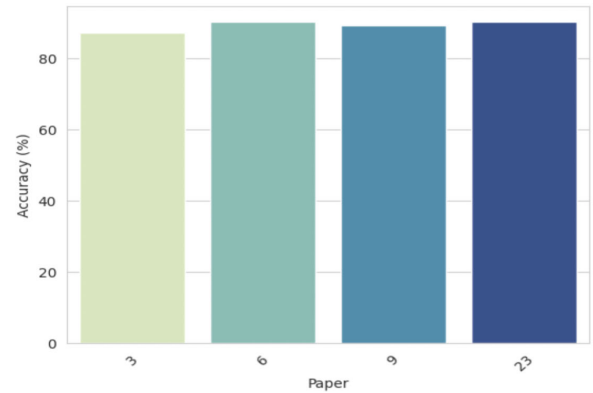


FIGURE 5. Compare the accuracy of different transformer based brain tumor classification algorithms, where the y-axis represents the accuracy and the x-axis represents the serial numbers from Table 2.

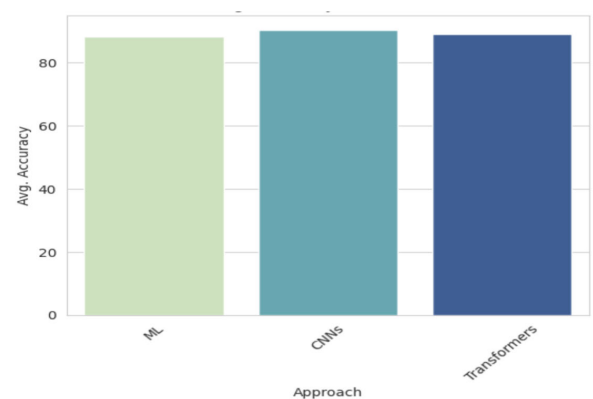


FIGURE 6. This figure shows the average accuracy of all approaches mentioned in Table 2. The y-axis represents the accuracy, while the x-axis corresponds to the serial numbers listed in Table 2.

TABLE 3. Comparative study between machine learning, CNNs, and transformers in brain tumor analysis.

Aspect	M.L.	CNNs	Transformers
Feature Representation and Extraction	No	Yes	Yes
Data Complexity and Scalability	No	Yes	Yes
Interpretability and Explainability	Yes	No	Yes
Performance and Accuracy	No	Yes	Yes
Data Efficiency and Generalization	No	Yes	Yes

networks (CNNs), and transformers. One significant gap identified is the need for further exploration of model robustness and generalization across diverse and heterogeneous datasets. While various studies have achieved high accuracy using different architectures and datasets, the robustness of these models in different imaging modalities and real-world clinical settings remains underexplored. Additionally, the integration of interpretability and explainability in model predictions is crucial for clinical acceptance but is often limited in current research. Furthermore, strategies such as transfer learning, domain adaptation, and multi-modal fusion have shown potential but require more comprehensive investigation to improve model performance and reliability, especially in scenarios with limited data. Addressing these

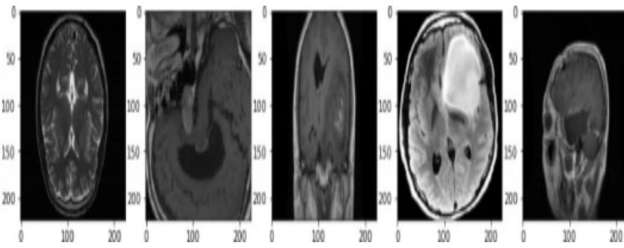


FIGURE 7. Visual depiction of multigrade brain tumor DataSet [107].

gaps can lead to more reliable, interpretable, and generalizable models for brain tumor detection, segmentation, and classification, ultimately contributing to better clinical decision-making and patient outcomes.

E. CHALLENGES IN BRAIN TUMOR TECHNIQUES

Despite the advancements in brain tumor assessment through machine learning (ML), convolutional neural networks (CNNs), and transformers, several challenges remain. One primary challenge is the variability and heterogeneity of medical imaging data, which can significantly impact model performance. Datasets used for training often differ in terms of acquisition protocols, resolutions, and patient demographics, leading to models that may not generalize well across diverse clinical environments. Another challenge is the need for large, annotated datasets, which are time-consuming and costly to produce. The complexity and black-box nature of deep learning models also pose a significant barrier to clinical adoption, as clinicians require interpretable and explainable results to trust and validate AI-driven decisions. Moreover, the robustness of these models against adversarial attacks and noise is a critical concern, especially in high-stakes medical applications where errors can have severe consequences. Lastly, the integration of multi-modal data, such as combining MRI and CT scans with clinical records, remains an underexplored area that could provide more comprehensive insights but introduces additional complexity in data processing and model design. Addressing these challenges is essential for the development of reliable, interpretable, and robust AI systems for brain tumor assessment.

IV. DATASET ANALYSIS AND PREPROCESSING

Brain tumor classification relies heavily on the availability and quality of datasets for training and testing machine learning models. These datasets, spanning different imaging modalities like MRI, CT, and histopathological images, are crucial for developing accurate and robust classification algorithms. This section, provide a analysis of key brain tumor classification datasets, highlighting their characteristics, strengths, limitations, and impact on medical imaging advancements.

A. DATASETS

This data contains publicly available datasets in brain tumor related research papers.

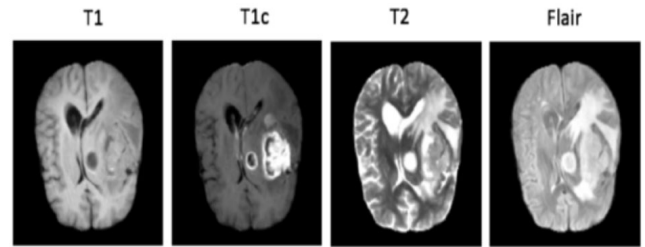


FIGURE 8. Visual depiction of BRATS-2019 brain tumor DataSet [8].

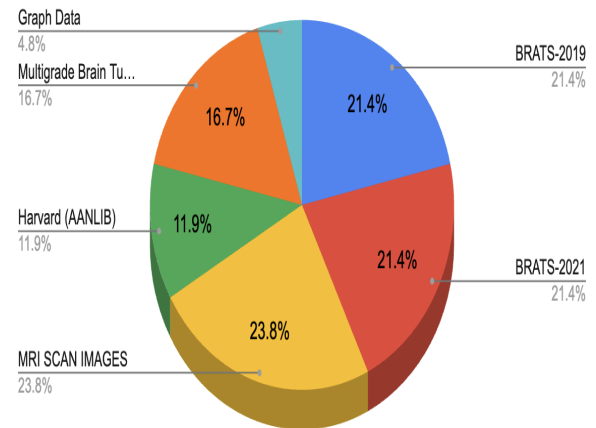


FIGURE 9. The pie chart above represents the dataset usage in Table 2.

1) MULTIGRADE BRAIN TUMOR DATA SET

This dataset [107] comprises two versions: the original dataset, comprising 121 MRI cases and an enhanced dataset that grows to 3630 MRI cases using the original pictures. Sample images are depicted in Fig. 8. Further details regarding data augmentation and other pertinent information are outlined within.

2) BRATS-2019

The BRATS-2019 [8] dataset is an extensive set of multi-modal MRI scans from throughout the world's medical centres that includes both preoperative and postoperative phases. It allows for a thorough evaluation of the characteristics of brain tumours and comprises. The provision of ground truth segmentation masks by expert annotations makes it easier to train and assess machine learning algorithms for automated tumour segmentation.

3) BRATS-2021

The BRATS-2021 [9] dataset continues the tradition of providing a comprehensive collection for the purpose of segmenting brain tumours from multimodal MRI data. It builds upon previous iterations by incorporating advancements in imaging technology and annotation techniques. The dataset provides a comprehensive perspective of the characteristics of brain tumours by combining MRI images from various sources.

4) HARVARD (AANLIB)

The Central Nervous System MRI images [10] are stored in an online repository called the whole-brain Atlas

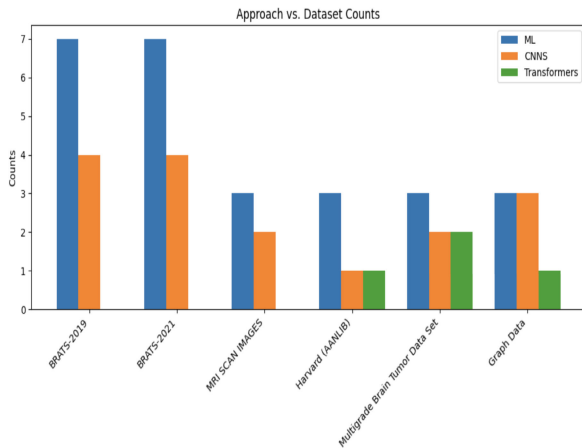


FIGURE 10. The bar chart above shows the approach-wise dataset count as detailed in Table 2.

(ANNLIB) [10]. This large, online database includes more than 13,000 brain MRIs from 30 different instances. This collection contains a wide variety of photos, such as tumour and normal scans, as well as examples of strokes, gliomas, Alzheimer's disease, and infectious disorders.

In reference to 2, the dataset utilized in our study corresponds to the Multigrade Brain Tumor Data Set [107]. This dataset includes contributions from various sources: BRATS-2019 [8], BRATS-2021 [9], Harvard (AANLIB) [10], MRI SCAN IMAGES [108]. And below is count per approach.

B. AUGMENTATION AND PREPROCESSING STEPS

To enhance the robustness and performance of machine learning models, it is crucial to apply augmentation and preprocessing techniques to the datasets [109]. The following are steps that are used in research papers:

Augmentation: - *Rotation*: Randomly rotating images to make models invariant to orientation. - *Flipping*: Applying horizontal and vertical flips to increase variability. - *Scaling*: Resizing images to introduce scale variance. - *Translation*: Shifting images horizontally and vertically. - *Brightness and Contrast Adjustment*: Randomly changing the brightness and contrast to simulate different lighting conditions. - *Noise Addition*: Adding Gaussian noise to simulate real-world noise.

Preprocessing: - *Normalization*: Scaling pixel values to a standard range (e.g., 0 to 1 or -1 to 1) to facilitate faster and more stable training. - *Resizing*: Ensuring all images are of the same dimensions, typically required for batch processing in neural networks. - *Histogram Equalization*: Enhancing the contrast of images to improve feature visibility. - *Masking*: Applying segmentation masks to focus on regions of interest, such as tumor areas. - *Cropping*: Removing irrelevant parts of the images to reduce computational load and focus on significant regions.

V. CHALLENGES

In BTc, one of the significant challenges is capturing non-linearity in the data. Because they are linear in design, traditional machine learning algorithms frequently fail to capture the intricate correlations found in medical imaging data. This limitation inhibits their ability to discern. Convolutional Neural Networks (CNNs) have partly addressed this challenge by inherently capturing non-linearities through their multiple layers of interconnected neurons. However, optimizing CNN architectures to effectively capture and utilize non-linear features remains an ongoing area of research.

Another challenge lies in ignoring the depth of tumors in 2D images. Conventional medical imaging techniques often produce two-dimensional representations of three-dimensional structures, such as brain tumors. Addressing this challenge requires developing methodologies that can effectively extract and incorporate depth-related features from 2D images or leveraging advanced imaging modalities that provide three-dimensional representations of tumors, such as volumetric MRI or CT scans.

Furthermore, there is a lack of multidimensional representation in existing brain tumor classification approaches. While 2D images provide valuable information, they may not capture the full spectrum of features relevant to tumor characterization. Tumors exhibit intricate spatial, temporal, and morphological characteristics that extend beyond what can be represented in a single image plane. Incorporating multi-dimensional representations, such as time-series data from dynamic imaging modalities or spatial information from multi-sequence MRI scans, presents a major obstacle because of the increased complexity of data processing and analysis.

Moreover, there is a need to experiment with attention mechanisms to enhance feature capture in brain tumor classification. Attention mechanisms allow neural networks to anchor on relevant regions of input data, potentially improving their ability to extract discriminative features for classification. However, integrating attention mechanisms into existing CNN or Transformer architectures for brain tumor analysis requires careful experimentation to determine optimal attention strategies and architectures. Additionally, understanding how attention mechanisms interact with other components of the classification pipeline and their impact on model interpretability is essential for developing robust and clinically relevant classification systems. By addressing these issues, techniques for classifying brain tumours will become more dependable and precise, which will eventually improve patient outcomes.

VI. RECOMMENDATIONS

To successfully overcome the difficulties in classifying brain tumours, several recommendations can be proposed:

- 1) **Integration of Non-linear Models:** Given the complexity of brain tumor data, integrating

TABLE 4. Comparison of challenges and recommendations in brain tumor classification.

Challenges	Recommendations
Non-linearity Capture	Integration of non-linear models such as deep neural networks (both Transformers and CNNs) can identify complex patterns and relationships in brain tumor data more effectively.
Ignoring Depth of Tumor in 2D Image	Utilization of three-dimensional imaging modalities (volumetric MRI or CT scans) to capture depth-related features and provide more comprehensive information for accurate tumor characterization.
Lack of Multi-dimensional Representation	Incorporation of multi-dimensional representations by integrating information from dynamic imaging modalities, multi-sequence MRI scans, and advanced imaging techniques to create more comprehensive tumor representations.
Experiment with Attention while Capturing Features	Experimentation with attention mechanisms such as self-attention in Transformers and spatial attention in CNNs to optimize feature extraction and improve classification performance by focusing on relevant regions of input data.

non-linear models, specifically CNNs and Transformers, can help capture patterns and relation in the data more effectively. Researchers should explore advanced architectures and optimization techniques to enhance the capacity of these models to capture non-linear features accurately.

- 2) **Utilization of Three-dimensional Imaging:** In order to extend the capabilities of imaging available today in order to address the issue identified above and other similar cases, researchers should aim at using tomographic scans. These modalities, when able to incorporate depth-related features into the classification functions, can deliver additional details that will aid in characterizing the tumors in the improved manner.
- 3) **Incorporation of Multi-dimensional Representations:** Rich spatial, temporal or morphological structures of tumors need to be accurately captured and hence, multi-dimensional representations are important for the classification pipeline. The authors suggest that further research should be conducted as to how data from dynamic imaging techniques, multi sequence MR scans or other advanced expansion techniques can be incorporated into planning so as to establish tumour models that are more versatile and useful.
- 4) **Experimentation with Attention Mechanisms:** By probing specific regions of the input, using attention mechanisms could benefit the ability of neural networks to focus its feature extraction on important areas of the image when used in brain

tumor classification. Scholars should also analyze different attention configurations like position-wise and channel-wise self-attention in transformers and spatial attention mechanisms in CNNs, with the purpose of enhancing the extraction of features and increasing accuracy of the models.

By implementing these recommendations, researchers can develop more robust and accurate methods for BTC, ultimately advancing and improving patient outcomes. Additionally, interdisciplinary collaboration between researchers, clinicians, and medical imaging experts is crucial for addressing these challenges effectively and translating research findings into clinical practice.

VII. CONCLUSION

In concluding our comprehensive survey on brain tumor classification techniques, we have traversed a landscape where medical imaging and machine learning intersect, witnessing the evolution from traditional methods to the emergence of CNNs and Transformers. Through comparative analyses, we have dissected the strengths and limitations of these methodologies, aiming to guide researchers in selecting optimal approaches for various diagnostic scenarios. Our exploration extended to the diverse landscape of datasets, equipping readers with insights necessary for informed decision-making. Addressing the challenges inherent in brain tumor classification, we have outlined strategies for overcoming hurdles and offered practical recommendations to drive future research.

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